

App Development Project Report

1. Student Details

Name: Ahmar Rehan

Roll Number: 24F2007062

Email: 24f2007062@ds.study.iitm.ac.in

About Me: I am pursuing the Bachelor of Science in Data Science from IIT Madras. I have a strong interest in full-stack web development, database management, and software engineering. Through this project, I gained practical experience in database design, role-based access control, Flask web development, and secure authentication.

2. Project Details

Project Title: Trekking Management Application

Problem Statement: Develop a centralized web application for managing trekking activities involving Admins, Trek Staff, and Trekkers, replacing manual coordination.

Approach: Built using Flask, SQLite, SQLAlchemy, Bootstrap, and Jinja2 with separate dashboards for Admin, Trek Staff, and Users.

3. AI / LLM Declaration

ChatGPT was used for understanding Flask concepts, debugging, session-based authentication, and documentation. The architecture, implementation, debugging, testing, and integration were completed and understood by me.

Estimated AI Usage: 15%

4. Technologies Used

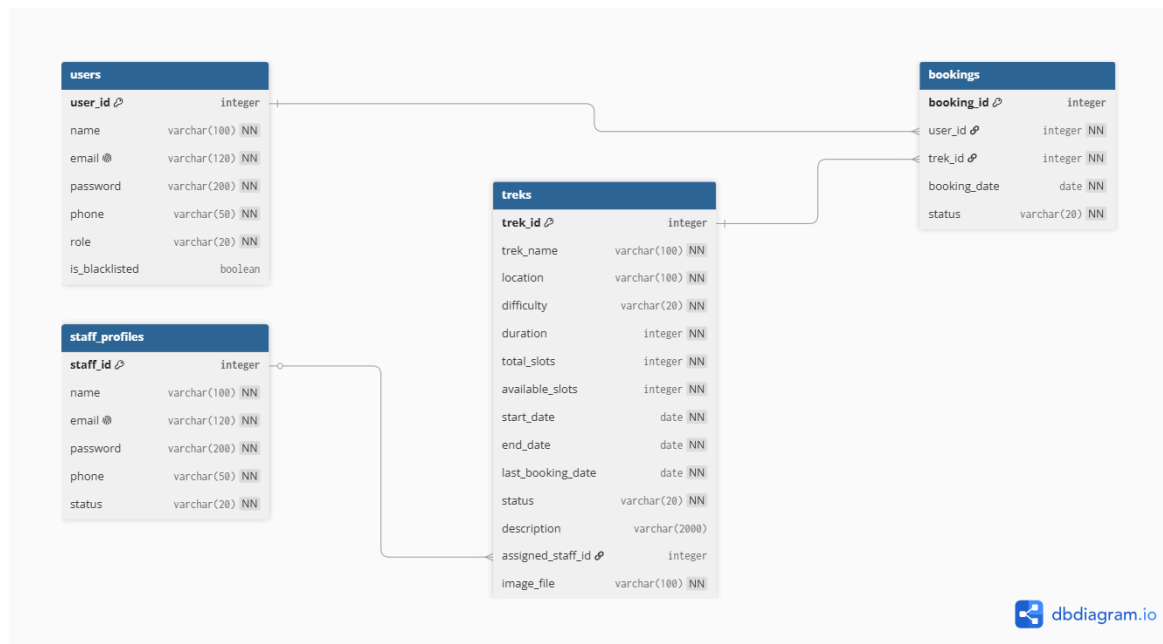
Flask, Flask-SQLAlchemy, SQLite, Jinja2, HTML5, CSS3, Flask Session, Werkzeug, Python. **Flask** is used as main backend, **Flask-SQLAlchemy** is used for database management, **SQLite** is used as a database, **Jinja2** used for templating and template inheritance, **HTML5** is used as primary frontend language and basic frontend validations, **vanilla CSS** is used for styling, **Flask Session** Authorization (Basic version).

App Development Project Report

5. Database Schema

Tables: User, Staff Profile, Trek, Booking.

Relationships: User→Booking(One-to-Many), Staff Profile→Trek(One-to-Many), Trek→Booking(Many-).



6. Features Implemented

Admin: Manage treks, users, staff, bookings, approvals and assignments.

Trek Staff: Manage assigned treks, update slots/status, manage participants.

User: Browse, search, filter, book and cancel treks, view booking history.

7. Security Features

Session-based authentication, role-based authorization, protected routes, URL tampering protection, logout using session clearing.

8. Project Structure

app.py, models.py security.py templates/ static/ instance/

9. Video Presentation

Google Drive Link: [Screen & Camera Recording - Jul 14, 2026 \(1\).mp4](https://drive.google.com/file/d/13ef9_rYQyKaXgJKtAlhEAtuNnzlaIQOq/view?usp=sharing) ,
https://drive.google.com/file/d/13ef9_rYQyKaXgJKtAlhEAtuNnzlaIQOq/view?usp=sharing